INCH-POUND

MIL-DTL-3970/13C
25 November 1998
SUPERSEDING
MIL-W-3970/13B
4 April 1977

DETAIL SPECIFICATION SHEET

# WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE H-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Complete requirements for acquiring the waveguide assemblies described herein shall consist of this document and the latest issue of MIL-DTL-3970.

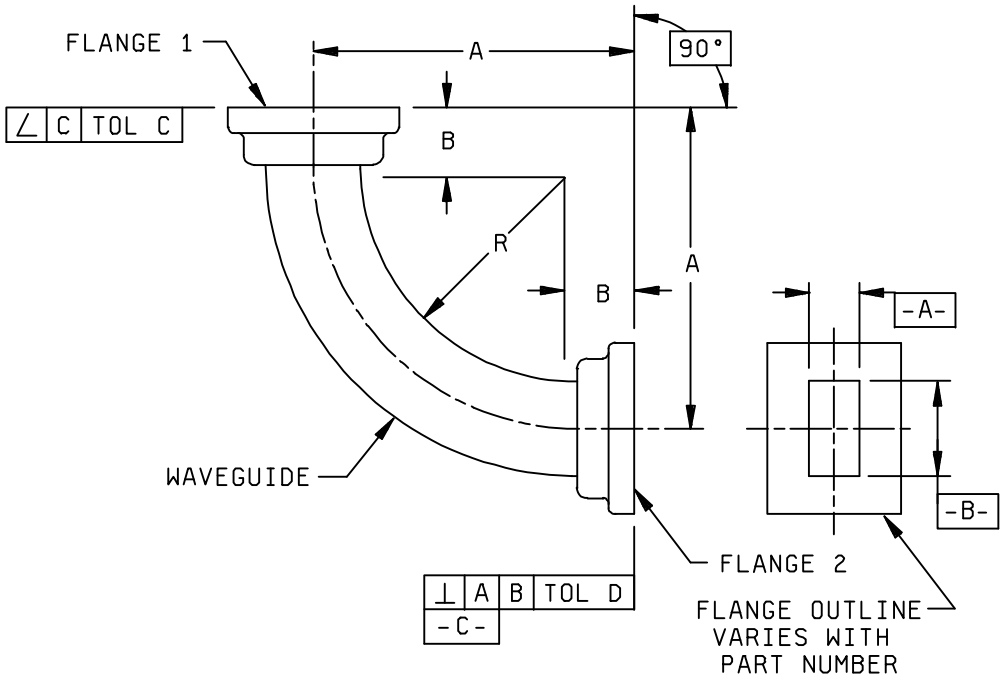


FIGURE 1. Assembly configuration.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:07Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, electrical and physical characteristics, and dimensions.

| Dash no. | Waveguide M85/ | Flange 1 M3922/ | Flange 2 M3922/ | Frequency range $TE_{10}$ mode | Insertion loss max | VSWR max | Pressurization $lb_f/in^2$ gage | Dimensions 1/ 2/ 3/ A | B nom | C | D | R |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | GHz | dB | | | | | | | |
| 001 | 1-043 | 56-001 | 61-002 | 2.6 - 3.95 | --- | 1.05 | 15 | 10.500 (266.70) ±.030 (±0.76) | 1.500 (38.10) | .115 (2.92) | .012 (0.30) | 7.500 (190.50) ±.125 (±3.18) |
| 002 | 1-055 | 57-002 | 62-002 | 3.95 - 5.85 | --- | 1.05 | 30 | 7.000 (177.80) ±.030 (±0.76) | 1.000 (25.40) | .077 (1.96) | .008 (0.20) | 5.000 (127.00) ±.125 (±3.18) |
| 003 | 1-067 | 55-001 | 60-001 | 5.85 - 8.20 | --- | 1.03 | 30 | 4.875 (123.83) ±.025 (±0.64) | .750 (19.05) | .056 (1.42) | .006 (0.15) | 3.375 (85.72) ±.125 (±3.18) |
| 004 | 1-073 | 53-002 | 59-007 | 7.05 - 10.0 | --- | 1.07 | 30 | 4.062 (103.17) ±.020 (±0.51) | .687 (17.45) | .044 (1.12) | .005 (0.13) | 2.750 (69.85) ±.063 (±1.60) |
| 005 | 1-079 | 53-001 | 59-006 | 8.20 - 12.4 | --- | 1.05 | 30 | 3.562 (90.47) ±.020 (±0.51) | .562 (14.27) | .038 (0.97) | .004 (0.10) | 2.500 (63.50) ±.063 (±1.60) |
| 008 | 1-077 | 53-003 | 59-008 | 8.2 - 12.4 | --- | 1.05 | 30 | 3.562 (90.47) ±.015 (±0.38) | .562 (14.27) | .038 (0.97) | .004 (0.10) | 2.500 (63.50) ±.063 (±1.60) |
| 009 | 1-071 | 53-004 | 59-009 | 7.05 - 10.0 | --- | 1.07 | 30 | 4.062 (103.17) ±.020 (±0.51) | .687 (17.45) | .044 (1.12) | .005 (0.13) | 2.750 (69.85) ±.063 (±1.60) |
| 010 | 1-041 | 56-002 | 61-001 | 2.60 - 3.95 | --- | 1.05 | 15 | 10.500 (266.70) ±.030 (±0.76) | 1.500 (38.10) | .115 (2.92) | .012 (0.30) | 7.500 (190.50) ±.125 (±3.18) |
| 011 | 1-089 | 53-005 | 59-001 | 12.4 - 18.0 | .07 | 1.04 | 30 | 2.601 (66.07) ±.015 (±0.38) | .500 (12.70) | .028 (0.71) | .003 (0.08) | 1.750 (44.45) ±.063 (±1.60) |
| 012 | 1-053 | 57-001 | 62-001 | 3.95 - 5.85 | --- | 1.05 | 30 | 7.000 (177.80) ±.030 (±0.76) | 1.000 (25.40) | .076 (1.93) | .008 (0.20) | 5.000 (127.00) ±.125 (±3.18) |
| 014 | 1-065 | 55-002 | 60-002 | 5.85 - 8.20 | --- | 1.03 | 30 | 4.875 (123.82) ±.025 (±0.64) | .750 (19.05) | .056 (1.42) | .006 (0.15) | 3.375 (85.72) ±.125 (±3.18) |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:07Z
Check the source to verify that this is the current version before use.

TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

| Dash no. | Wave-guide M85/ | Flange 1 M3922/ | Flange 2 M3922/ | Frequency range $TE_{10}$ mode | Insertion loss max | VSWR max | Pressur-ization $lb_f/in^2$ gage | Dimensions 1/ 2/ 3/ A | B nom | C | D | R |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | GHz | dB | | | | | | | |
| 015 | 1-093 | 53-005 | 59-001 | 12.4 - 18.0 | .07 | 1.04 | 30 | 2.601 (66.07) ±.015 (±0.38) | .500 (12.70) | .028 (0.71) | .003 (0.08) | 1.750 (44.45) ±.063 (±1.60) |
| 017 | 1-018 | 52-002 | 52-002 | 1.12 - 1.70 | --- | 1.05 | 30 | 7.810 (198.37) ±.010 (±0.25) | 1.116 (28.35) | .106 (2.69) | .028 (0.71) | 3.364 (85.45) ±.125 (±3.18) |
| 031 | 1-018 | 52-002 | 52-002 | 1.12 - 1.70 | .10 | 1.05 | 1.5 | 14.000 (355.60) ±.015 (±0.38) | 2.000 (50.80) | .160 (4.06) | .029 (0.74) | 8.670 (220.22) ±.015 (±0.38) |
| 032 | 1-029 | 52-006 | 52-006 | 1.70 - 2.60 | .10 | 1.05 | 3.0 | 10.000 (254.00) ±.015 (±0.38) | 2.000 (50.80) | .101 (2.57) | .019 (0.48) | 5.770 (146.56) ±.015 (±0.38) |

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part or Identifying Number (PIN): M3970/13-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

| PIN M3970/13- | Type designation | PIN M3970/13- | Type designation |
|---|---|---|---|
| 001 | UG-712/U | 009 | UG-720/U |
| 002 | UG-713/U | 010 | UG-721/U |
| 003 | UG-714/U | 011 | UG-722/U |
| 004 | UG-715/U | 012 | UG-723/U |
| 005 | UG-716/U | 013 | UG-724/U |
| 006 | UG-717/U | 014 | UG-725/U |
| 007 | UG-718/U | 015 | UG-726/U |
| 008 | UG-719/U | 016 | UG-727/U |

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

| PIN M3970/13- | Manufacturer CAGE code | Commercial type |
|---|---|---|
| 017 | 10001 | 2938971-1 |
| 017 | 01456 | 650BH49 |
| 017 | 05865 | L2257 |
| 017 | 17250 | L11114 |
| 031 | 52564 | 3705001 |
| 032 | 52564 | 4005001 |

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 11, 17, 99

Preparing activity:
DLA - CC

(Project 5985-1118 )



Source: https://assist.dla.mil -- Downloaded: 2026-10-17T03:07Z
Check the source to verify that this is the current version before use.